DAIO Defense AI Observatory

Guarding the Alliance's Eastern Frontiers

Defense AI in Poland

Rafał Lipka, Jakub Wernik, and Wojciech Dzięgiel

DAIO Study 25|29

Ein Projekt im Rahmen von

dtec.bw
Zentrum für Digitalisierungs- und Technologieforschung der Bundeswehr

Finanziert von der Europäischen Union
NextGenerationEU

About the Defense AI Observatory

The Defense AI Observatory (DAIO) at the Helmut Schmidt University in Hamburg monitors and analyzes the use of artificial intelligence by armed forces. DAIO comprises three interrelated work streams:

- Culture, concept development, and organizational transformation in the context of military innovation
- Current and future conflict pictures, conflict dynamics, and operational experience, especially related to the use of emerging technologies
- Defense industrial dynamics with a particular focus on the impact of emerging technologies on the nature and character of techno-industrial ecosystems

DAIO is an integral element of GhostPlay, a capability and technology development project for concept-driven and AI-enhanced defense decision-making in support of fast-paced defense operations. GhostPlay is funded by the Center for Digital and Technology Research of the German Bundeswehr (dtec.bw). dtec.bw is funded by the European Union – NextGenerationEU.

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Content

1 Summary

Poland has recently launched an ambitious plan to integrate artificial intelligence (AI) technologies into its defense sector as part of a broader digital transformation agenda. These initiatives reflect both the need to enhance the effectiveness of public institutions and the determination to modernize the Polish Armed Forces in line with evolving security challenges. The Ministry of National Defense has assumed a central role in this process, with growing emphasis on cross-ministerial collaboration and institutional support.

A milestone in this effort has been the publication of Poland's Military AI Strategy 2024–2039 by the Ministry of National Defense and the establishment of the Artificial Intelligence Implementation Center within the Cyberspace Defense Forces. This decision reflects a more centralized and coordinated approach, designed to capitalize on existing expertise and ensure synergies between the armed forces, industry, and the research sector. At the same time, significant challenges remain, particularly in securing adequate funding and building the necessary human capital to sustain these initiatives.

Although the adoption of AI in the military is still in its early stages, Poland's efforts mirror the approach taken by leading NATO nations. Strategic documents adopted by the Ministry of National Defense outline a broad research and development agenda, covering areas such as Cognitive AI, Data Science, Cybersecurity, Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance (C4ISR), Human-Machine Interfaces (HMI), and autonomous platforms, among others. In addition to identifying specific R&D priorities, the government has also launched national programs—such as PERUN—designed to stimulate the development of domestic solutions in these fields.

At the same time, several AI technologies have already been developed and deployed by the Cyberspace Defense Forces Component Command, particularly in the area of cybersecurity, where they are used to enhance threat detection. Also notable in this context is the Legion Battle Management System, currently being developed by specialists within the Cyberspace Defense Forces. Its AI-enabled architecture reportedly allows for the automated analysis of imagery data provided by unmanned platforms operated by the Polish Armed Forces. Furthermore, AI is increasingly being integrated into operational systems, including unmanned aerial vehicles.

To date, little is known about the allocation of funds for acquiring AI-enabled systems for the Armed Forces, as such expenses are embedded within the broader procurement programs of the Ministry of National Defense. Nonetheless, significant resources have been dedicated to establishing the Artificial Intelligence Implementation Center, which is tasked with coordinating most efforts related to the development of AI solutions for the military. At the same time, the government is preparing additional funding mechanisms—such as the Artificial Intelligence Fund and the Defense and Security Fund—intended to support research and development of dual-use technologies.

AI is emerging as a key tool for enhancing training within the Polish Armed Forces, with potential applications ranging from personalized e-learning and automated assessment to advanced simulations and VR-based scenario design. While not a substitute for live exercises, AI-enabled simulators can reduce both the costs and risks of training, while strengthening decision-making skills at all levels of command. The growing role of AI is also reshaping military education, with institutions such as the Military University of Technology in Warsaw integrating AI into curricula and expanding cooperation with industry. At the same time, the Ministry of National Defense is establishing partnerships with global technology companies—including Microsoft and Oracle—to modernize training, strengthen cybersecurity, and build cloud and AI infrastructure, ensuring long-term benefits for both defense and the broader economy.

Poland's defense-focused AI development stands in contrast to the broader national AI ecosystem, which remains fragmented and dominated by multinational corporations. Although domestic enterprises and startups are pursuing AI solutions with both civilian and military applications, the country continues to rely heavily on foreign technology providers. Without targeted policies and sufficient resources to support domestic innovation and strengthen public-private cooperation, this dependency is likely to persist for the foreseeable future.

The defense industry—especially state-owned enterprises—faces additional structural challenges. Despite increasing demand for advanced technologies, these companies remain heavily dependent on public procurement and are largely absent from international markets. As Poland acquires major weapon systems from foreign contractors, the need to automate production processes and integrate AI into manufacturing becomes increasingly urgent, particularly given the country's demographic constraints.

Poland continues to align its defense innovation agenda with broader trends among NATO allies. Initiatives such as the Cyber LEGION program and strong performance in international exercises like Locked Shields demonstrate the country's growing competence in cyber operations and its willingness to involve civilian specialists in national defense. This also illustrates a pragmatic approach that seeks to balance military requirements with the expertise of the wider technological community in Poland.

While Poland's defense AI strategy remains in the early stages of implementation, the direction taken by Polish decision-makers is promising. Success will depend on the government's ability to translate broad strategies into concrete policies, secure sustained investment, and foster closer collaboration between the armed forces, industry, and academia. If these conditions are met, Poland could emerge as a regional leader in military AI integration and contribute significantly to strengthening NATO's eastern flank and the Alliance's broader defense capabilities.

2 Thinking About Defense AI

2.1 Defense and Security Landscape – Enormous Needs, Even Greater Challenges Ahead

The geostrategic situation is undoubtedly key to understanding Poland's stance on the implementation of AI-based technologies in the broader context of military modernization efforts and the state's overall security policy. Poland's current approach to security and defense can best be summarized by the ancient Latin adage attributed to Vegetius, a Roman writer from the 4th century CE: "If you want peace, prepare for war."

After decades of neglect, the Polish government was compelled to rapidly rethink its priorities following Russia's invasion of Ukraine in February 2022 and, consequently, to expand the nation's military power through the large-scale acquisition of a wide range of weapon systems. With a defense budget projected to reach approximately PLN187bn (USD50bn)[1] or 4.7 percent of GDP in 2025,[2] more than half of which is allocated to defense procurement,[3] Poland continues to build up its military with the ultimate goal of becoming the most capable land power in Europe.[4] Nonetheless, Poland's military shopping spree is not limited to heavy land-based equipment such as main battle tanks, artillery systems, missile launchers, and various types of armored vehicles. It also includes the acquisition of state-of-the-art combat aircraft, air defense systems, unmanned aerial systems (UAS), a wide range of naval vessels, and numerous other advanced combat systems across all branches of the Polish Armed Forces.[5]

Many of these weapon systems either already incorporate or are designed to be integrated with AI technologies to varying degrees, especially in the areas of real-time data processing, autonomous operation, and threat detection. All of this is accompanied by structural reforms, personnel expansion, and a multitude of new initiatives aimed at enhancing Poland's defense capabilities and deterrence posture. The modernization and transformation effort is not only a response to conventional military threats but also to a broader strategic shift toward an intense, data-centric battlespace, in which decision-making requires greater automation. Consequently, AI is emerging as a strategic enabler and an essential component of the armed forces, allowing for optimization of human and material resources as well as rapid reactions to counter emerging threats.

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1 Based on the average USD/PLN exchange rate in early June 2025. The Polish 2025 defense spending is expected to reach approximately PLN 187 bln in 2025.
2 "Odpowiedzialny, ale hojny – budżet na 2025 rok przyjęty" [Responsible but Generous – 2025 Budget Adopted].
3 Dmitruk, "Wydatki na obronność Polski w latach 2022-2025 według MON" [Poland's defense spending in 2022–2025 according to the Ministry of National Defense].
4 Mader, "Poland on the way to becoming the strongest army in Europe."
5 Lipka/Czub, "Si vis pacem, para bellum: Transformation of the Polish Armed Forces in Response to Emerging Threats."

The threat lurking in the East is not purely military in nature, as Poland is also facing numerous hybrid threats that challenge the security of its infrastructure, energy supply, cyberspace, and even its physical borders. These broader security challenges also create a need for AI-enhanced solutions.

Border security is a case in point as hostile policies of Minsk and Moscow have created constant migratory pressure in recent years. The governments of Belarus and Russia have been orchestrating an influx of migrants from less developed nations, exploiting the vulnerability of these people and weaponizing their plight against Poland, the Baltic states, and the entire European Union with the aim of creating chaos and political instability. As a result, in late 2021, the Polish government opted for the construction of a physical barrier along the Polish-Belarusian border,[6] which is accompanied by electronic surveillance systems and other security measures that Poland continues to develop to date in order to reduce the probability of illegal incursions into Polish territory.[7] In April 2023, Poland's Ministry of the Interior and Administration announced the implementation of similar measures in northeast Poland, along the 200 km-long Polish-Russian border.[8]

These migration-related issues should be viewed within the broader European context of a continent-wide migrant crisis that has impacted numerous countries, including Poland's key ally, neighbor, and most important economic partner—Germany. In autumn 2024, the government of the Federal Republic introduced checks at all land borders, affecting the movement of people and goods between Poland and Germany.[9] Rising tensions between the two capitals, coupled with fears of uncontrolled, illegal migration from Germany to Poland, led Warsaw to implement similar measures in July 2025.[10]

The internal disorder triggered by migratory pressures highlights the vital importance of initiatives aimed at strengthening the security of the EU's external borders—not only for the European Union and Poland, but also for NATO, given the hybrid warfare tactics employed by Russia and Belarus. These tactics have included deliberate acts of sabotage targeting infrastructure and businesses in Poland. For instance, in March 2025, Polish prosecutors charged a Belarusian citizen with orchestrating an arson attack that destroyed a large retail store in Warsaw.[11] Reports indicate that the individual acted in cooperation with Russian intelligence and special services.[12]

6 Drabik, "Poland: Fence in the swamp."
7 "Poland Effectively Protects the EU Border."
8 "Ruszyła budowa zapory elektronicznej na granicy polsko-rosyjskiej" [Construction of the electronic barrier on the Polish-Russian border has begun].
9 "Checks at all German land borders start on Monday."
10 Doerries/Barth, "Poland imposes checks on German and Lithuanian borders amid migration fears."
11 "Poland charges Belarusian with arson on behalf of Russian intelligence."
12 Ibid.

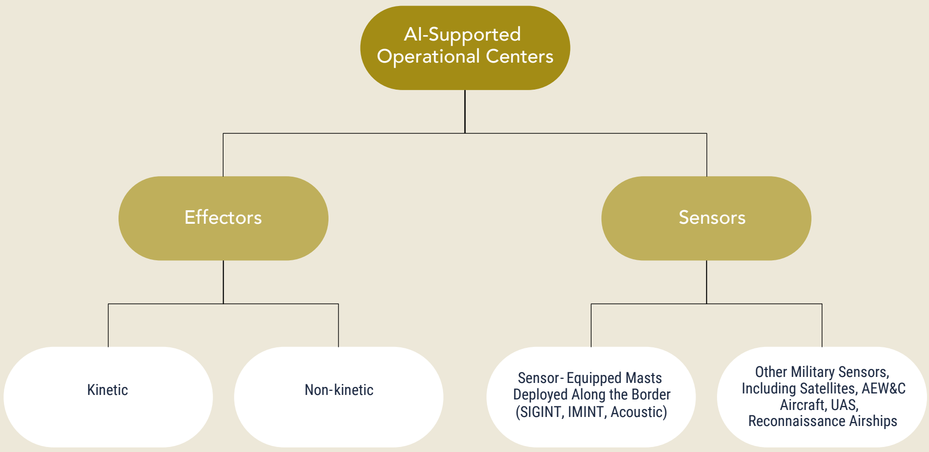
A series of arsons across Poland between early and mid-2025, affecting businesses, residential buildings, and critical infrastructure, raises serious concerns about the security situation on NATO's eastern flank and the potential for ongoing hybrid attacks by the Russian Federation.[13] While these incidents may not appear directly linked to the development and deployment of AI in the defense sector, their scale and frequency have clearly influenced the thinking of Polish policymakers, prompting them to accelerate efforts to strengthen national security and the protection of external borders.

In May 2024, the Polish government announced a new initiative codenamed "East Shield," aimed at enhancing Poland's security and border defense from a strictly military perspective.[14] The National Deterrence and Defense Program – East Shield is arguably one of the most comprehensive efforts of its kind in recent Polish history. Its ultimate goal is to establish a system of fortifications, bunkers, trenches, minefields, anti-tank barriers, and other critical infrastructure, all supported by advanced surveillance systems.[15] The government emphasizes the defensive nature of this initiative and hopes that such a system will serve as a powerful deterrent against potential Russian aggression. The strategic importance of the program for the EU's overall defense capabilities was also recognized in the White Paper for European Defense – Readiness 2030, published by the European Commission in March 2025.[16]

It is worth noting that effective border protection requires the integration of various types of sensors, which generate vast amounts of data necessitating real-time analysis and rapid decision-making regarding appropriate countermeasures based on the nature of the threat. These demands are among the key reasons for the planned acquisition of advanced data management systems that make extensive use of AI (Figure 1). Such systems have reportedly already been deployed to support the physical barrier along the Polish-Russian[17] and the Polish-Belarusian border,[18] with further implementation of even more sophisticated solutions planned under the East Shield program.[19]

13 Krzysztoszek, "Series of fires across Poland, Russian sabotage suspected."
14 "Tarcza Wschód" wzmocni bezpieczeństwo Polski i wschodniej flanki NATO" ['Eastern Shield' will strengthen the security of Poland and NATO's eastern flank].
15 "Narodowy Program Odstraszania i Obrony – Tarcza Wschód" [National Deterrence and Defence Programme – East Shield].
16 "White Paper for European Defence – Readiness 2030."
17 Bujak, Tomasz, Interview by Graf, Jędrzej, "Polska sztuczna inteligencja dla Tarczy Wschód" [Polish artificial intelligence for the East Shield].
18 Mężyński, Andrzej, "Sztuczna inteligencja na granicy z Białorusią. MSWiA zmodernizowało "zaporę techniczną" [Artificial intelligence on the border with Belarus: The Ministry of the Interior and Administration has modernized the 'technical barrier'].
19 These solutions will include the deployment of drone detection and tracking systems, incorporating radars, thermal imaging, and acoustic monitoring. These capabilities will strengthen the ability to counter unmanned aerial vehicles, which have become increasingly prevalent in recent conflicts, as demonstrated by the war in Ukraine. https://tarczawschod.wp.mil.pl/en/about-the-programme/ (last accessed 5 September 2025).

Figure 1: Overview of the Selected Components of the East Shield Program[20]



2.2 Strategic Approach to the Implementation of Defense AI

At the strategic level, Poland's national AI development strategy is based on the 2020 Policy for the Development of Artificial Intelligence in Poland, the implementation of which is a consequence of the publication of numerous documents related to Poland's digitalization and development efforts. These include the Public Data Opening Program, the Strategy for Responsible Development, the From Paper to Digital Poland Program, the Dynamic Poland 2020 Strategy for Innovation and Efficiency of the Economy, among others.[21] The strategy also takes into account the Coordinated Plan on Artificial Intelligence of the European Commission, as well as other strategic recommendations and AI-related guidelines published by the European Union and the Organization for Economic Co-operation and Development (OECD).

20 Based on the Presentation of the General Staff regarding the implementation of the East Shield Program. https://tarczawschod.wp.mil.pl/en/about-the-programme/ (last accessed 5 September 2025).
21 "Policy for the Development of Artificial Intelligence in Poland from 2020. Appendix to the Resolution no. 196 of the Council of Ministers of 28 December 2020 (item 23)."

The policy deliberately excludes the defense and security dimensions of the state's activities, while emphasizing that the civilian sector is expected to cooperate with the military in multiple areas, as outlined in the 2020 National Security Strategy of the Republic of Poland. These include the need for development and implementation of new, highly innovative technologies, including "automated and robotized weapon platforms using artificial intelligence."[22] However, the policy does reference the military and other key actors, including the Ministry of National Defense, as major future users of AI-based solutions. For example, it identifies the Ministry of National Defense as one of the key stakeholders in advancing AI development in Poland, particularly in areas such as education, scientific research, and collaboration with highly innovative enterprises. The strategy also emphasizes the need to increase national investment in AI for defense and security purposes.[23]

Given the deterioration of the security situation on NATO's eastern flank, a new National Security Strategy is highly anticipated. The updated strategy will likely address key challenges and outline development priorities across multiple areas, including those related to AI. As of late July 2025, it remains unclear when the final version of the document will be published, although the new Strategy was officially approved by the government on July 25, 2025.[24] It is worth noting, however, that the President's Chancellery drafted its recommendations for the upcoming strategy in July 2024, identifying the development of AI and autonomous systems as top priorities, particularly in the field of education and research and development (R&D).[25]

In the defense context, the most important document concerning artificial intelligence is Poland's Ministerial Artificial Intelligence Strategy until 2039 (*Resortowa strategia sztucznej inteligencji do roku 2039*)—hereafter referred to as *Poland's Military AI Strategy 2024–2039*—published in August 2024 by the Ministry of National Defense. The Strategy's primary objective is to develop, integrate, and implement AI in the areas of defense and security under the Ministry's jurisdiction. This includes the creation of dedicated structures responsible for:

1. Coordinating all AI-related initiatives at the Ministry level;
2. Conducting research and development, as well as testing and implementing selected solutions provided by partner institutions;
3. Maintaining and operating deployed systems.[26]

22 National Security Strategy Of The Republic Of Poland.
23 Ibid.
24 "Strategia Bezpieczeństwa Narodowego Rzeczypospolitej Polskiej" [National Security Strategy of the Republic of Poland].
25 "Rekomendacje do Strategii Bezpieczeństwa Narodowego Rzeczypospolitej Polskiej" [Recommendations for the National Security Strategy of the Republic of Poland].
26 "Resortowa strategia sztucznej inteligencji do roku 2039" [Poland's Ministerial Artificial Intelligence Strategy until 2039].

The document presents a holistic approach to defense-related applications of AI by highlighting the strategic context for the development of such technologies, analyzing the Ministry's current level of readiness for AI implementation and outlining a vision for the advancement of military AI in Poland. Moreover, it is assessing opportunities and threats, identifying key future directions and objectives for the Ministry, and recommending actions that need to be taken. The Strategy explicitly mentions the role of AI for data analytics to support decision making, for instance in the context of targeting and intelligence, surveillance and reconnaissance (ISR) operations. Advancements in AI technologies are also emphasized in the context of improving cybersecurity and developing autonomous weapon systems, including the deployment of drone swarms. The strategy further highlights the use of such technologies in information warfare, while noting that AI can pose a significant threat in the hands of non-state actors, including terrorist organizations, as well as state actors that may use it to conduct various forms of hybrid operations, such as disinformation campaigns.

According to the authors, the implementation of advanced technologies—including AI—is a prerequisite for maintaining capable Armed Forces with a strong deterrence posture. By 2039, AI-based technologies are expected to be widely deployed in the following areas:

1. Autonomous weapon systems used in both offensive and defensive operations. These systems will be capable of operating fully autonomously or supporting operators in navigation or decision-making, thereby enhancing operational effectiveness and reducing risks to military personnel.
2. Data collection and analytics in intelligence, surveillance, and reconnaissance (ISR) operations. AI will enable the effective processing of large volumes of data to anticipate and assess adversary actions, and to support planning and execution of military operations. The automation is expected to significantly reduce the workload of military personnel, as well as increase the effectiveness of target detection and identification.
3. Military logistics, including planning, resource management, transportation, maintenance, repair and operations (MRO), and supply chain management. The implementation of AI technologies is expected to significantly reduce risks related to human error and improve cost-effectiveness of logistics operations.
4. Cyber operations, by increasing the effectiveness of both defensive and offensive actions in cyberspace.
5. Personnel training, through more efficient, adaptive, personalized and data-driven approaches.
6. Military decision-making, particularly in enhancing data analytics and situational awareness to support tactical, operational and strategic-level judgments by providing accurate assessments and recommendations.

Besides threats such as hostile cyber operations against military AI systems as well as potential malfunctions and errors, among others, the strategy highlights the importance of legal and moral aspects of developing and deploying AI-based solutions for the military purpose.

Developing AI, automation and autonomous systems is a key topic of the *Research Priorities of the Ministry of National Defense for 2021–2035* (*Priorytetowe kierunki badań naukowych w resorcie obrony narodowej w latach 2021-2035*), introduced by Ministerial Decision No. 2/DIn on 10 January 2023.[27] This document has undoubtedly influenced the final shape of Poland's Military AI Strategy. Its main objective is to identify the critical technologies that must be developed and implemented within the Polish Armed Forces to ensure the country possesses the necessary defensive capabilities by the mid-2030s. The document places the development of AI and related technologies at the top of the Ministry's extensive list of R&D priorities. The authors highlight several potential applications of AI, including its integration into Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance (C4ISR) systems. AI is also envisioned as a key enabler in training programs and simulators using virtual and augmented reality (VR/AR) technologies, offering a cost-effective way to enhance soldiers' combat readiness and improve competencies related to maintaining advanced weapon systems. Other applications include, among others, space technologies such as satellite systems, and innovations in combat medicine.

The broad vision for AI implementation in the Polish military—though vague in some areas or left unspecified due to the classified nature of many applications—is best summarized by the words of the Polish Minister of National Defense, Władysław Kosiniak-Kamysz:

> We are not only building cybersecurity, but we also aim to stay ahead of our adversaries and competitors, while simultaneously becoming a leader of an alliance of nations of good will – countries that allocate their spending toward innovative and effective solutions. We are a NATO record-holder in terms of spending on transformation. Transformation is not just about purchasing equipment – it is an investment in digital security and the implementation of artificial intelligence. In a few years, there will be no military capable of functioning without artificial intelligence. We are fully aware of this – perhaps more so than some other countries in this part of Europe and the world. We know that without the implementation of artificial intelligence, it will be impossible to train soldiers, make logistical decisions, conduct operations, or manage the battlefield.[28]

27 "Priorytetowe kierunki badań naukowych w resorcie obrony narodowej w latach 2021-2035" [Priority directions of scientific research in the Ministry of National Defense for 2021–2035].
28 "Wicepremier W. Kosiniak – Kamysz: nie ma dziś ważniejszej sprawy niż bezpieczeństwo Polaków" [Deputy Prime Minister W. Kosiniak-Kamysz: There is no more important issue today than the security of Poles].

Last but not least, it is worth noting that the application of AI is one of the key components of Poland's highly anticipated Digitalization Strategy 2035. Based on the publicly available draft, the Strategy is unlikely to address the defense-related aspects of AI implementation, as its primary focus lies on the economic and industrial implications of adopting advanced digital technologies. Nonetheless, the Strategy clearly emphasizes the importance of applying AI in areas such as cybersecurity and public administration, while also encouraging collaboration between the civilian and military sectors due to the dual-use nature of these technologies.[29]

2.3 Preparing for the Implementation of AI for Defense and Security Purposes

Government documents—particularly Poland's Military AI Strategy 2024–2039—place strong emphasis on the readiness of the Ministry of National Defense and the Armed Forces to deploy various technologies utilizing AI, as outlined in the *Operational Goal no 1: Preparation for fielding artificial intelligence in the Ministry*. The Ministry's approach is largely based on the concept of human-machine teaming (HMT) "in order to multiply the existing capabilities, combining human's cognitive abilities and creativity with machines' analytical capabilities and speed."[30] This highlights the supportive role of AI within the Ministry and the Polish Armed Forces, with humans remaining the ultimate decision-makers and bearing full responsibility for the consequences of employing such technologies.

Paradoxically, one of the greatest challenges facing the government in this area is the availability of a skilled civilian workforce and military personnel—including both the competencies of current Ministry staff and the government's ability to attract highly skilled and knowledgeable talent. The Strategy addresses these issues by presenting a set of actions aimed at enhancing personnel competencies across technical, operational, ethical, and legal dimensions of AI implementation. Regarding the latter, the Ministry has recognized the need to review existing legislation and operational procedures to ensure they are adapted for the integration of AI—particularly in situations where certain aspects of command and control (C2) may be delegated to such systems. The document also outlines measures to retain specialists and recruit staff with the necessary skills and expertise, recommending that the employment practices be aligned with those applied in the field of cybersecurity. It is worth noting that, for the time being, the Ministry of National Defense can capitalize on the growing interest in voluntary military service and

29 "Strategia cyfryzacji państwa" [National Digitalization Strategy].
30 "Resortowa strategia sztucznej inteligencji do roku 2039" [Poland's Ministerial Artificial Intelligence Strategy until 2039].

the public's heightened engagement with defense and security matters driven by the ongoing conflict in Ukraine. Over the past two years, the Ministry has seen a significant increase in applications, prompting the government to increase yearly recruitment limits.[31] However, it remains unclear whether these developments will positively impact efforts to recruit specialists in the field of AI.[32]

Given the ambitious plans to deploy AI in multiple domains simultaneously, the Ministry of National Defense will also need to update its ICT (Information and Communication Technology) infrastructure to ensure adequate computing power for both current and future needs. This includes adopting cloud-based services, developing classified on-premise systems, and establishing new procedures to safeguard data security. The latter is particularly important as the development of AI solutions for the Ministry will involve research institutes and private companies that will require access to highly sensitive and classified information. This will pose a significant challenge in the near future, especially given that the anonymized and secure databases required for developing specific AI systems reportedly do not exist.[33]

Significant attention has been given to the organizational preparedness of the Ministry and the Armed Forces. The Ministry has drafted a plan to establish dedicated departments within its organizational structure, including the General Staff, to effectively coordinate AI-related projects—particularly those requiring collaboration with other ministries and international partners. Simultaneously, the Strategy introduces the concept of creating an AI Implementation Center, which would be responsible for the technical and technological aspects of deploying AI in the defense sector, as well as for coordinating R&D efforts in this field.

Although the Strategy repeatedly emphasizes the importance of adhering to national and international law—including International Humanitarian Law—as well as ethical norms in the operation of AI-enabled military systems, it provides little detail on specific considerations in this regard. The document explicitly acknowledges ethical challenges related to delegating decision-making authority to combat systems, particularly in the use of kinetic measures to destroy selected targets. Nonetheless, it reaffirms that the development and implementation of AI in the military will comply with the provisions of the 2023 Political Declaration on Responsible Military Use of Artificial Intelligence and Autonomy, which Poland

31 "MON zwiększa na ten rok limit dla ochotników w DZSW i obniża w 2025 r." [The Ministry of National Defense increases the limit for volunteers in the Voluntary Basic Military Service for this year and lowers it in 2025].

32 On 13 August 2025, MG Karol Molenda stated: "Throughout Poland, under the patronage of the Cyber Command, we established uniformed IT classes. We also increased the number of students at military universities. We focused on long-term development. Now we can see tangible results. Today, I had the opportunity to welcome over 130 second lieutenants, graduates of cybersecurity-related programs, to our ranks." Quoted based on: https://x.com/MolendaKarol/status/1955696958833557603 (last accessed 5 September 2025).

33 "O sztucznej inteligencji na seminarium naukowym w CDIS SZ" [On Artificial Intelligence at the Scientific Seminar in the CDIS SZ].

ultimately endorsed[34] following cross-ministerial consultations—despite the government's initial reluctance to sign due to the reportedly unclear implications of endorsement.[35]

It is also important to note that Poland has yet to enact a dedicated AI law that would fully align national regulations with the EU Artificial Intelligence Act of 2024, although existing legislation is being gradually adapted.[36] A draft of Poland's AI Law was published in late 2024 for public consultation, which concluded in February 2025.[37] As of August 2025, a revised draft has been released, providing for the establishment of an AI Development and Security Authority.[38]

34 "Political Declaration on Responsible Military Use of Artificial Intelligence and Autonomy."
35 Palczewski, "Wojsko: deklaracja o użyciu AI. Brakuje Polski; MSZ wyjaśnia." [Military: Declaration on the Use of AI. Poland Missing; Foreign Ministry Explains].
36 Dudkowiak, "AI Law in Poland."
37 "Projekt Ustawy o systemach sztucznej inteligencji ze zmianami po konsultacjach społecznych" [Draft Act on Artificial Intelligence Systems with Amendments Following Public Consultations].
38 Ciechoński, "Rządowe prace nad ustawą wdrażającą AI Act na ostatniej prose" [Government Work on the Law Implementing the AI Act in the Final Stretch].

3 Developing Defense AI

The Ministry's concept of developing and implementing AI is primarily based on the scalability of adopted solutions, as outlined in the *Operational Goal no 2: Scalable Implementation of Artificial Intelligence*.[39] Given the innovative and rapidly evolving nature of AI technologies—as well as the need for collaboration between the Ministry and various entities, including other ministries and the industry—a learning-by-doing approach is considered optimal in the context of changing capabilities and potential future applications. The authors place significant emphasis on the need to establish procedures for testing, verification, and validation of machine learning algorithms, which will require the active involvement of developers. Considering these factors, the Ministry intends to promote a systematic, yet incremental and evolutionary approach to developing new solutions through experimentation with less sophisticated models. This hybrid approach—combining on-premise systems with commercial AI solutions, such as those utilizing cloud architecture—is expected to effectively stimulate the growth of AI innovation within the defense sector. Furthermore, the government plans to engage all key stakeholders in this process, including developers, procurement entities, and future system operators. This will involve applying DevSecOps (development, security, and operations) principles throughout the development and implementation lifecycle.

Poland's efforts to accelerate the AI implementation process build on the experience gained from the establishment of the Cyberspace Defense Forces in 2022, following the Minister's decision issued in 2019.[40] Moreover, the aforementioned Artificial Intelligence Implementation Center—whose formation began in 2025—will operate within the structure of the Cyberspace Defense Forces Component Command, which is capable of ensuring access to specialists with required expertise, at least during the Center's initial phase of operations.[41] Its primary objective will be to implement AI technologies in key areas of military operations, including:

- Analysis of intelligence and reconnaissance data,
- Development of autonomous combat systems,
- Decision-making support and logistics optimization,
- Verification and implementation of available AI technologies for military applications.

Given the unit's capabilities to conduct the full spectrum of cyber operations and its ability to attract highly qualified talent, the government's plans appear well-founded and appropriately tailored to Poland's needs, while significantly

39 "Resortowa strategia sztucznej inteligencji do roku 2039" [Poland's Ministerial Artificial Intelligence Strategy until 2039].
40 "2022 – rok, w którym powstały Wojska Obrony Cyberprzestrzeni" [2022 – the year in which the Cyberspace Defense Forces were established].
41 Based on the data provided by the Cyberspace Defense Forces Component Command, following the authors' request for information.

reducing the risks associated with establishing such an organization within the armed forces. This also demonstrates the awareness of Polish decision-makers regarding recent trends among leading NATO member states in the fields of digitalization and AI. For example, dedicated AI development and implementation centers have been established in countries such as the United States,[42] the United Kingdom,[43] and many others.[44]

3.1 R&D Priorities

Prior to the publication of Poland's Military AI Strategy 2024–2039—which provides a primary road map outlining the government's policy on developing and adopting AI in the Polish Armed Forces and the Ministry of National Defense—the National Center for Research and Development launched the PERUN program in November 2023.[45] The main objective of this initiative is to support development of innovative projects through collaboration between the industry, research institutes, and academia in the field of defense and security, as outlined in the *Research Priorities of the Ministry of National Defense for 2021–2035*. Noteworthy is the fact that six of the program's thematic areas are directly connected to the development of AI-based technologies and the automation of defense systems. These include applications in cybersecurity, intelligence, surveillance, and reconnaissance, drone swarms, and the so-called manned-unmanned teaming, which aims to enhance the effectiveness of cooperation between human-operated platforms and autonomous systems.[46] Expected outcomes of these projects include, among others:

1. The implementation of AI in automating security-related processes within ICT systems;
2. Countering disinformation through the analysis of digital data to detect Deepfake content;
3. Automation of ISR data analysis;
4. The development of AI-enabled communication protocols for drone swarms operating in coordination with manned systems.

42 "Chief Digital & Artificial Intelligence Office Celebrates First Year"
43 For more, see: https://www.gov.uk/government/groups/defence-artificial-intelligence-centre (last accessed 3 September 2025).
44 Borchert, "The Very Long Game of Defense AI Adoption: Introduction," pp. 18-20.
45 "Czas na realizację projektów w zakresie badań naukowych i prac rozwojowych na rzecz obronności i bezpieczeństwa państwa – ruszył Program NCBR pk. PERUN" [Time to implement projects in the field of scientific research and development work for national defense and security – the NCBR program code-named PERUN has been launched].
46 "Załącznik nr 2 do Regulaminu Konkursu 1/PERUN/2023" [Appendix No. 2 to the Rules of Competition 1/PERUN/2023].

The implementation of the PERUN program reflects an evolutionary and incremental approach to advancing innovative defense technologies in Poland. This marks a considerable improvement over the previous program of such kind, codenamed SZAFIR,[47] which ended in 2022 and was considered rather outdated, failing to account for rapid technological advancements and insights drawn from the ongoing conflict in Ukraine.

It is important to emphasize that the need for a state-funded program aimed at accelerating the development of domestic high-tech innovations arises from limited R&D funding opportunities in Poland, a highly competitive international environment, and the constraints imposed by similar international initiatives, all of which present serious challenges to many Polish companies. This is particularly relevant in the case of Polish state-owned defense enterprises, which are heavily dependent on public procurement and—with very few exceptions, such as Mesko, the producer of the GROM/PIORUN man-portable air-defense systems[48]—remain virtually absent from international markets due to competition from much larger and more technologically advanced counterparts.[49] Therefore, the establishment of such programs is essential to stimulate the development of domestic solutions, thereby enhancing the industry's competitiveness and its capacity to generate innovations in the long term.

It is worth noting that almost 200 projects have been positively evaluated under the formal requirements of the PERUN program. Given the classified nature of these endeavors, it is currently difficult to determine the exact number of submitted AI-related applications. However, the range of applicants—comprising both leaders and members of participating consortia—includes universities, R&D institutions, and private as well as state-owned companies, with a significant number of applications submitted by public military universities acting as project leaders.[50]

While the PERUN program is intended to support various R&D projects through 2033, its scope and content do not fully reflect the strategic, long-term research priorities of the Ministry of National Defense. These long-term plans for the development and implementation of AI are outlined in the *Research Priorities of the Ministry of National Defense for 2021–2035*. The most important components of this plan, along with their anticipated impact on the functioning of the Polish Armed Forces, are presented in Table 1.

47 "4/SZAFIR/2021."
48 "PIORUN MANPADS."
49 Compared to data for 2022, the value of actual exports and intra-EU transfers of arms and military equipment in 2023 increased by over €572M, reaching €1,753M. In terms of value, Ukrainian entities ranked first for recipients of products exported from Poland in 2023 (approximately 81% of export value), followed by the United States (7%), Spain, and Germany (1% each). For more, see: https://www.gov.pl/web/dyplomacja/eksport-uzbrojenia-i-sprzetu-wojskowego-z-polski-raport-za-rok-2023 (last accessed 5 September 2025).
50 "Lista wniosków pozytywnie ocenionych pod względem formalnym (Konkurs nr 1/PERUN/2023)" [List of Applications Positively Evaluated in Terms of Formal Criteria (Competition No. 1/PERUN/2023)].

Table 1: Selected Research Priorities of the Ministry of National Defense for 2021–2035 in the field of AI[51]

Field of Research	Impact on the Armed Forces	Approach to R&D Financing	Due Date
1. Cognitive AI	■ Establishing efficient mechanisms for machine learning ■ Developing new algorithms and procedures for effective decision-making ■ Creating automated software solutions for detecting and countering cyber threats	National and international R&D programs	2035
2. Data Science	■ Using big data analytics to uncover hidden relationships between data sets ■ Improving the efficiency of identifying solutions to operational needs and challenges	National and international R&D programs	2033
2.1. Command and Control	■ Supporting command and control processes ■ Streamlining logistics ■ Reducing overall operational costs	National and international R&D programs	2035
2.2. Cybersecurity	■ Safeguarding operations in the cyber domain ■ Enhancing the efficiency of cyber threat detection ■ Automating countermeasures to cyber threats ■ Ensuring the effectiveness of both offensive and defensive cyber operations ■ Strengthening the capacity to develop new solutions based on collected data and operational experience	National R&D programs	2035
2.3. ISR (Intelligence, Surveillance, and Reconnaissance)	■ Reducing the time required for imagery data analysis ■ Increasing the effectiveness of threat detection systems ■ Enabling content analysis of audio and text data to enhance system capabilities	National R&D programs	2030

51 Based on: "Priorytetowe kierunki badań naukowych w resorcie obrony narodowej w latach 2021-2035" [Priority directions of scientific research in the Ministry of National Defense for 2021–2035].

Field of Research	Impact on the Armed Forces	Approach to R&D Financing	Due Date
3. Development of an Efficient Human-Machine Interface (HMI)	■ Improving situational awareness for commanders overseeing units composed of both soldiers and unmanned platforms ■ Enabling effective human-machine cooperation ■ Enhancing the ergonomics of systems used to control unmanned platforms	Programs of the National Center for Research and Development	2028
4. Development of Autonomous Platforms	■ Reducing operational risks related to human error ■ Increasing the efficiency of logistics and medical evacuation from combat zones ■ Enhancing reconnaissance capabilities ■ Improving the safety of military personnel during operations	Programs of the National Center for Research and Development and through international consortia within the frameworks of the European Defense Agency (EDA) and the European Defense Fund (EDF)	2028

Table 1 demonstrates that most of Poland's R&D priorities related to artificial intelligence were projected to rely primarily on national sources of funding, although participation in international frameworks such as the European Defense Agency and the European Defense Fund is also anticipated. It is worth noting that Polish state-owned companies have faced significant challenges in the past when applying for dedicated EU funds. For example, under the Act in Support of Ammunition Production (ASAP), only one Polish company—Dezamet—succeeded in securing modest funding to expand its manufacturing base.[52]

Although Polish entities are generally interested in participating in EDF-funded projects, most past participants have been research and development institutes,

52 Król, "MON wyjaśnia aferę z grantami na amunicję" [Defense Ministry Addresses Ammunition Grant Scandal].

universities, and private companies involved in large consortia as subcontractors.[53] The absence of Polish companies in consortia conducting AI-related R&D projects within the EDF framework in 2024 is concerning. However, it is worth noting that two Polish entities—WB Electronics and the Military University of Technology—are taking part in the 2023 Artificial Intelligence Deployable Agent (AIDA) project of the European Defense Fund.[54] This highlights both the growing awareness among Polish companies of the opportunities related to EDF's support for innovative projects and the capacity of the private sector and academia to successfully secure such funding.

3.2 Overview of Poland's Defense AI Ecosystem

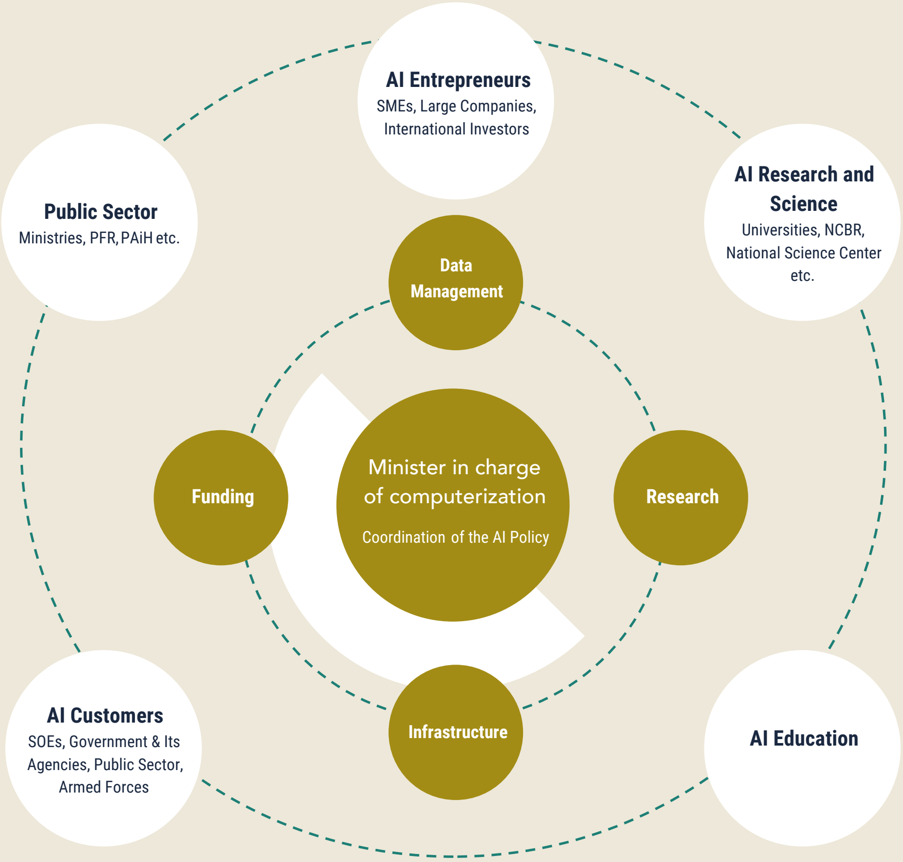
Current efforts to develop AI-based solutions in Poland appear to focus primarily on Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance (C4ISR), as well as the automation of related processes—although other innovations, such as the development of AI-supported autonomous systems, are also being pursued. The broader AI ecosystem (Figure 1) remains relatively diverse, encompassing both defense companies and firms focused on dual-use technologies, along with startups and other entities established to stimulate the growth of such innovations in Poland. Collaboration between state-owned companies within the Polish Armaments Group (PGZ, Polska Grupa Zbrojeniowa) and startups has also been observed—suggesting that the development and implementation of AI in the defense and security sectors may provide emerging Polish businesses with new opportunities.

The Ministry of National Defense's perspective on the development of Poland's AI ecosystem is best captured by Brig. Gen. Marcin Górka, head of the Ministry's Innovation Department:

> Our goal is to build an innovation ecosystem that responds to the military's needs and supports technology development in collaboration with the defense industry. Stability and inclusiveness are important to us – the Ministry of Defense's doors are open to civilian technologies that can be adapted to enhance national security. The IDA program demonstrates that we are ready to collaborate with innovators and that we want them to better understand the needs of the armed forces.[55]

53 Palowski, "Miliard z UE na technologie obronne. Skorzysta polski przemysł?" [A Billion from the EU for Defense Technologies. Will Polish Industry Benefit?].
54 "Artificial Intelligence Deployable Agent."
55 "IDA DemoDay 2024: Polskie technologie dual-use dla bezpieczeństwa i rozwoju sektora obronnego" [IDA Bootcamp 2025 – PFR seeks innovative solutions for defense].

Figure 2: **Poland's AI Ecosystem According to the Policy for the Development of Ai in Poland from 2020**



Companies

This section will concentrate primarily on Polish entities or on enterprises with a strong connection to the Polish market—whether through their founders' backgrounds, their origins, or the scale of their operations in Poland. Nonetheless, it is important to note that the Polish government cooperates with numerous multinational technology companies and other global entities that undeniably shape the overall picture of the Polish defense AI ecosystem. Such cooperation agreements have been signed to date with Google,[56] Microsoft,[57] Oracle,[58] and Intel.[59] AI-based services with potential applications in administration and other public services are also promoted by major consulting firms, including Accenture.[60]

Other notable examples include international defense contractors that not only integrate AI into their combat platforms[61] but also transform the manufacturing of defense systems by implementing smart factory technologies.[62] These examples illustrate why mapping a complete picture of the AI ecosystem is so challenging: an ever-growing number of companies are adopting such technologies to varying degrees across a wide range of applications. This trend, in turn, directly influences the perspective of Polish decision-makers and local entities, which must contend with fierce competition from multinational companies firmly established in global markets. At the same time, it also creates opportunities for new frameworks of cooperation that could benefit the Polish Armed Forces and support the transformation of the domestic defense industry through well-designed procurement programs and offset agreements.

A key example of a domestically developed solution is the Legion Battle Management System (BMS), currently being introduced into operational service. It was developed by the previously mentioned Cyberspace Defense Forces Component Command, a specialized unit of the Polish Armed Forces responsible for cybersecurity and AI implementation.[63] Additionally, AI has reportedly been integrated into the EyeQ Air and EyeQ Land systems, developed by WB Group—one of Poland's most successful private defense companies, known particularly for its unmanned aerial systems. These systems are designed to automate the processes of target identification, classification, tracking, and threat assessment, thereby significantly enhancing operators' situational awareness while reducing their workload. They are reportedly ready for integration with fire control and battle management systems operated by the Polish Armed Forces.[64] It is also worth noting that WB Group is said to be the only Polish company participating in the AIDA (Artificial Intelligence Deployable Agent) project, which is funded by the European Defence Fund.[65]

56 "Google Invests Billions in Poland's Digital Future."
57 "Microsoft Invests PLN 3 Billion in a New Data Center in Poland."
58 "Współpraca DKWOC z Oracle w obszarze cyberbezpieczeństwa" [DKWOC's Cooperation with Oracle in the Field of Cybersecurity].
59 "AI for Industry – rozwój kompetencji AI w polskim przemyśle. Porozumienie MRiT-Intel podpisane" [AI for Industry – Developing AI Competencies in Polish Industry. MRiT–Intel Agreement Signed].
60 "Accenture zasilane generatywną sztuczną inteligencją" [Accenture Powered by Generative Artificial Intelligence].
61 "Already in the global defense industry, unmanned and artificial intelligence (AI) are becoming essential elements of weapons development."
62 "Smart Factories Transition from Automation to Intelligence."
63 "SI BMS Legion."
64 "AI Systems."
65 "WB GROUP in the AIDA project."

The AI race has attracted numerous startups that either fully specialize in developing such solutions or operate at the intersection of AI and other technologies. Poland reflects this trend, with various entities either originating in the country or choosing to establish offices in major Polish cities to strengthen their presence in the European market. As a result, the startup scene in Poland appears particularly dynamic, especially considering that AI is classified as a dual-use technology—offering both a high return on investment and diverse applications across military and civilian sectors.[66]

One such company is Digica, which maintains offices in Germany, the United Kingdom, the United States, and Poland. The company provides a wide range of AI-based services across multiple sectors, including healthcare and defense. Its solutions are reportedly being implemented in several NATO member states. In 2024, Digica was awarded a contract by PIT-RADWAR, a subsidiary of the Polish Armaments Group, to develop AI solutions aimed at enhancing the capabilities of PIT-RADWAR's hardware, including its radar systems.[67]

DataWalk is another company with Polish origins, which currently runs its offices in both Poland and the United States. The company portrays itself as an alternative to the famous American company Palantir, and even if it is less known than its US-based competitor, the company has achieved numerous successes by receiving contracts for the development of solutions for the U.S. Department of Justice Money Laundering and Asset Recovery Section (MLARS), U.S. Department of Defense, U.S. Department of State, as well as numerous disclosed and undisclosed businesses and public institutions from Europe, Asia, Africa, and Americas.[68] The company specializes in "revealing patterns, relationships, and anomalies for large-scale, multi-source intelligence investigations and data analysis", with a potential application in supporting decision making in the military.[69]

AI and machine learning are also among the key technologies used in the SKYctrl anti-drone system developed by the Polish company Advanced Protection Systems (APS).[70] The SKYctrl system integrates, among other components, 3D MIMO radars, tracking and identification systems supported by AI, as well as jammers, and is reportedly ready for integration with hard-kill systems. It is worth noting that the system has been battle-tested in Ukraine as early as 2022.[71] This milestone has contributed to the adoption of the company's products not only by Poland but also by the armed forces of Saudi Arabia, the UAE, and Qatar.[72] Although the

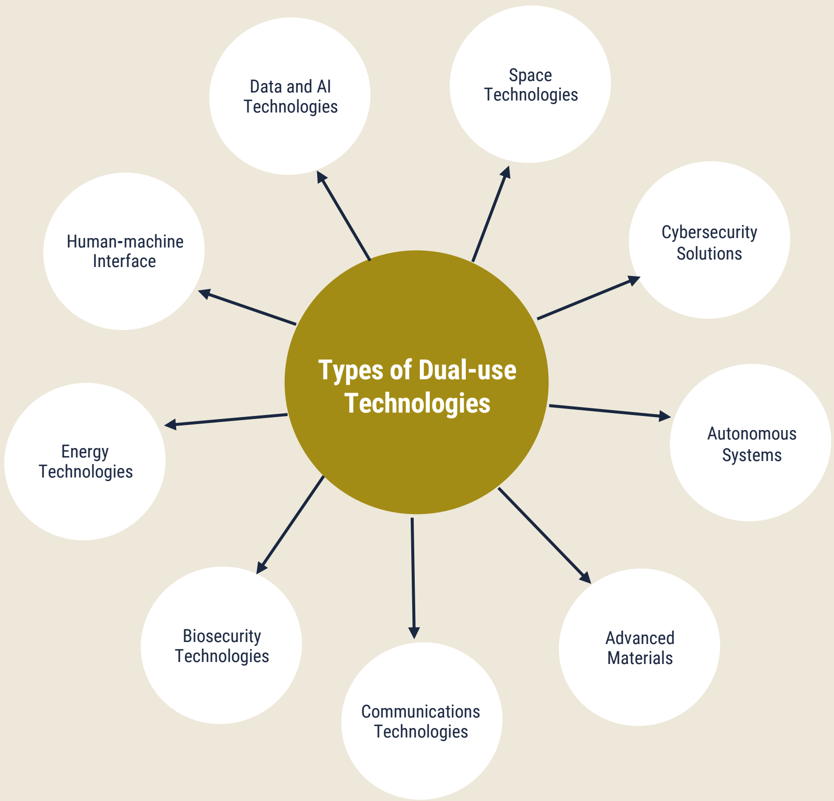
66 "Navigating Dual-Use Technologies: A Founder's Handbook to Opportunities and Challenges."
67 "PIT-RADWAR S.A. podpisał umowę z firmą Digica" [PIT-RADWAR S.A. signed a contract with Digica].
68 "For more, see: https://datawalk.com/company/our-customers (last accessed 5 September 2025).
69 For more, see: https://datawalk.com/company (last accessed 5 September 2025).
70 "SKYctrl Comprehensive anti-drone system."
71 Kushnikov, "The Ukrainian military received the SKY CTRL anti-drone system."
72 "FIELDctrl Ultra precise 3D MIMO radars."

military application of SKYctrl has garnered the most attention, the system can also be deployed to protect various types of critical infrastructure, such as airports, power plants, and government buildings, with many additional potential use cases where hostile drones may pose a threat.[73]

The development and implementation of AI in the defense and security sectors is not solely the domain of relatively new companies and startups. A growing number of well-established Polish tech firms are also investing in such technologies. For instance, the Polish company TELBUD, founded in 1987, has developed and successfully commercialized a family of PSIM-class (Physical Security Information Management) systems under the ARGUS brand.[74] One publicly disclosed application of this technology is the integration of various protection and security systems along the Polish-Russian and Polish-Belarusian borders. The system has also been adopted by the Polish Air Navigation Services Agency, which is responsible for managing air traffic in Poland.[75] The list of companies developing such solutions in Poland is considerably longer and likely includes numerous well-established technology firms and startups offering a variety of products and services at different stages of development and commercialization. However, due to limited access to information, the examples mentioned above should be regarded as reference points for the Polish defense AI ecosystem rather than as a complete list of businesses active in this field. It is also important to note that certain AI technologies currently under development may have dual-use applications in the future, and a much broader market analysis would therefore be required to fully assess the potential of domestic firms in this regard. Given that a growing number of specialized defense companies in Poland are likely to join this trend in the coming years—particularly with the development of new technologies and solutions by relatively small firms capable of partnering with both private and state-owned enterprises—the emergence of these players highlights the need for a better understanding of other non-business entities that contribute to the sector's growth.

73 For more, see: https://apsystems.tech/en/markets (last accessed 5 September 2025).
74 "For more, see: https://telbud.pl/en/about-us/our-history (last accessed 5 September 2025).
75 Bujak, Tomasz, Interview by Graf, Jędrzej, "Polska sztuczna inteligencja dla Tarczy Wschód" [Polish artificial intelligence for the East Shield].

Figure 3: Types of Dual Use Technologies[76]



Research and Technology Organizations

In the context of Poland's defense R&D ecosystem, it is crucial to highlight the role of the National Center for Research and Development (Narodowe Centrum Badań i Rozwoju, NCBR), an executive agency of the Polish government operating under the direct supervision of the Minister of Science and Higher Education. The NCBR is responsible for fostering cooperation between scientific institutions and businesses, as well as providing funding for relevant R&D projects, including those related to defense and national security—such as the previously mentioned PERUN program.

76 Based on: https://startup.pfr.pl/artykul/premiera-przewodnika-dla-specjalistow-z-obszaru-dual-use-navigating-dual-use-technologies (last accessed 5 September 2025).

The Łukasiewicz Research Network, which encompasses more than 20 specialized research institutes across Poland, is a prominent example of a public entity established to stimulate innovation and technological advancement in the country. Research and development in defense and national security is identified as a top priority among the various areas in which the network operates.[77] In mid-2025, the Łukasiewicz – AI Institute was officially inaugurated in Katowice, Silesia, as a specialized branch of the network responsible for coordinating R&D projects in AI and cybersecurity.[78] Although the institute is expected to focus primarily on implementing AI in the broader economy while supporting the digitalization of Polish small and medium-sized enterprises (SMEs), its creation demonstrates that public institutions recognize the importance of fostering the AI ecosystem and driving business transformation in Poland.

Additional research activities address the development of Explainable AI, aimed at creating algorithms whose decision-making processes are fully transparent and easily understandable to human users.[79] Researchers at the Military University of Technology in Warsaw (WAT) are currently working on implementing such solutions in medicine to enhance diagnostics, which could increase the efficiency of the healthcare system and potentially enable further advances in combat medicine. These technologies could not only have a tangible impact across a broad range of medical fields but also improve the reliability of AI-based solutions and strengthen trust in them.[80] Other research projects conducted by WAT scientists include AI algorithms for analyzing satellite imagery,[81] designing road traffic management solutions,[82] creating AI-supported techniques to detect deepfake content,[83] and building automated cybersecurity solutions aimed at improving threat detection efficiency.[84]

Two new institutes will be key for future research and development on defense AI. First, the Council of Ministers under the joint supervision of the Minister of Digitalization and the Minister of National Defense have launched the IDEAS Institute in February 2025.[85] The Institute will conduct research in AI, with a primary focus on applications in healthcare, defense, and public administration. However, as noted during the legislative process, the legal status of a research institute entails certain

77 For more, see: https://lukasiewicz.gov.pl/en/get-to-know-us (last accessed 5 September 2025).
78 "Łukasiewicz – AI oficjalnie otwarty. Rozwój sztucznej inteligencji i cyfryzacja biznesu to nasze priorytety" [Łukasiewicz – AI officially opened. The development of artificial intelligence and business digitalization are our priorities].
79 Naruszko,"Sztuczna inteligencja skróci kolejki do lekarza?" [Will Artificial Intelligence Shorten Wait Times for Doctors?].
80 Ibid.
81 Wrzos, "Naukowcy z WAT opracowali algorytm do oceny wysokości budynków" [Scientists from WAT Have Developed an Algorithm for Assessing Building Heights].
82 Wrzos, "Sztuczna inteligencja zredukuje korki na drogach" [Artificial Intelligence Will Reduce Traffic Jams].
83 Ołowniuk, "Deep fake – jak rozpoznać fałszywe treści?" [Deepfake – How to Recognize Fake Content?].
84 "Wojskowa Akademia Techniczna stworzyła narzędzie AI do wykrywania cyberataków" [The Military University of Technology Has Developed an AI Tool for Detecting Cyberattacks].
85 "Rozporządzenie Rady Ministrów z dnia 25 lutego 2025 r. w sprawie utworzenia Instytutu Badawczego IDEAS" [Regulation of the Council of Ministers of February 25, 2025, on the Establishment of the IDEAS Research Institute].

risks and operational challenges. In addition, the level of public funding allocated for its first year poses a substantial obstacle.

Second, the Artificial Intelligence Implementation Center (CISI) has been tasked to coordinate R&D projects, facilitating collaboration between the military, academia, and industry, and, finally, evaluating commercial products and services available on the market to advance defense AI.[86] According to Maj. Gen. Karol Molenda, commander of the Polish Cyber Command, the Center is not intended to function as another R&D hub; instead, it will focus primarily on the development and practical application of technologies created by other entities, which should considerably shorten the time needed to bring such solutions into operational service.[87] The Center will also integrate data from various manned and unmanned combat platforms of the Polish Armed Forces to enhance operational awareness.[88]

The establishment of the Center could potentially address one of the key challenges related to conducting R&D projects in Poland. Although R&D spending in Poland remains relatively low compared to other leading European countries—for instance, in 2023 Poland allocated only 1.56 percent of GDP to R&D, while countries such as Sweden, Belgium, Austria, Germany, and Finland spent over 3 percent—the main obstacle lies in the efficient use of funding and the commercialization of scientific projects.[89] The structural problems contributing to this situation are beyond the scope of this report; however, it is important to note that the existence of the Center could significantly increase the proportion of projects that are successfully implemented rather than ending up among the many solutions that never find practical application.

Additional Ecosystem Elements

It is noteworthy that Poland is also actively involved in allied initiatives such as DIANA (Defence Innovation Accelerator for the North Atlantic). In 2024, Kraków Technology Park (KPT) and the AGH University of Science and Technology announced the establishment of the FORT Kraków, which will operate in close collaboration with the Innovations Department of the Ministry of National Defense.[90] The creation of the FORT Kraków marks a significant milestone in the development of Poland's AI ecosystem, given its focus on supporting startups specializing in

86 "W Wojsku Polskim powstaje Centrum Implementacji Sztucznej Inteligencji" [An Artificial Intelligence Implementation Center is being established in the Polish Armed Forces].
87 Molenda, Karol, Interview by Okulska, Inez, "Piąty żywioł w starciu z AI" [The Fifth Element in Confrontation with AI].
88 "Gen. Molenda: 14 kwietnia rusza Centrum Implementacji Sztucznej Inteligencji" [Gen. Molenda: The Artificial Intelligence Implementation Center will launch on April 14].
89 "Gross domestic expenditure on research and development in 2023."
90 "Kraków DIANA – rusza polski akcelerator innowacji obronnych NATO" [Kraków DIANA – Launch of the Polish NATO defense innovation accelerator].

innovative dual-use technologies—including AI—that could substantially enhance the defense capabilities and resilience of NATO member states.[91]

These institutions will provide startups with business development expertise and technological acceleration, leveraging AGH's advanced R&D infrastructure. In addition to the accelerator site in Kraków, Poland hosts seven dedicated DIANA test centers—including the Center for Geospatial and Satellite Analysis (CENAGIS), the Industrial Research Institute for Automation and Measurements PIAP (Łukasiewicz – PIAP), Łukasiewicz – EMAG, and the Poznań Supercomputing and Networking Center (PSNC)—available to innovators participating in the DIANA program. These sites offer not only the infrastructure required to evaluate new technologies but also the specialized expertise of their personnel.[92] For instance, PSNC operates an aerospace laboratory featuring a hangar, research facilities, and a training ground for testing technologies such as unmanned aerial and ground vehicles, AI applications, as well as cybersecurity and quantum computing solutions. Łukasiewicz PIAP, on the other hand, specializes in AI-driven robotics and unmanned systems, among other areas. Besides the institutions mentioned above, it is also worth emphasizing the role of organizations such as the Military Institute of Engineer Technology (Wojskowy Instytut Techniki Inżynieryjnej, WITI), the Military Institute of Armored and Automotive Technology (Wojskowy Instytut Techniki Pancernej i Samochodowej, WITPiS), and the Military University of Technology, which will serve as key end-users of AI technologies. All institutions within the DIANA network bring significant experience in supporting such initiatives and maintain diverse networks of business and institutional partners, including both Polish and international defense companies and public agencies.[93]

Initiatives such as DIANA are also complemented by programs launched by other institutions, including the Polish Development Fund (Polski Fundusz Rozwoju, PFR), a state-owned company focused on providing financial and expert assistance to Polish entrepreneurs, as well as investing in large-scale development projects. One such initiative is the Investments Dual-use Adaptation (IDA) program, which aims to accelerate the growth of domestic innovations.

The program enables participating innovators to assess whether their solutions could have dual-use applications and to obtain expert guidance on how to advance their technology to meet military requirements. Through the program, companies gain direct access to potential end-users of their technologies, including the military, defense contractors, defense experts, and representatives of the Ministry of National Defense. Beyond its technical aspects, the program also

91 "About Krakow DIANA Accelerator."
92 "DIANA is partnering with more than 180 test centres from across the NATO Alliance."
93 "Testing Services for DIANA Startups: Capabilities of 7 Polish Test Centers in NATO DIANA Network."

emphasizes the value of networking, offering participants opportunities to connect with potential investors, such as representatives of venture capital (VC) funds.[94] The 2025 edition of the IDA program is supported by Palantir, Iceye, WB Group, and Polish Armaments Group, among others.[95]

The construction of Poland's first AI factories—Cyfronet AGH in Kraków and PIAST-AI in Poznań—marks an essential and long-awaited investment in the national AI ecosystem. PIAST-AI is designed to serve as an innovation hub, providing the services and infrastructure needed to drive advancements in IT and cybersecurity, healthcare and life sciences, the public sector, space technologies, robotics, and cross-sector sustainable development.[96] At the same time, the Ministry of Digital Affairs is pursuing participation in the European Commission's initiative to establish five AI Gigafactories across the EU. Under this program, the Ministry plans an investment of PLN5bn (approximately €1.15bn), including PLN2bn (around €0.47bn) from the state budget, to acquire 30,000 graphics processing units (GPUs) for a cluster of five leading research centers in Poland by 2029.[97]

Preliminary Insights from Ukraine

The battlefields of Ukraine offer valuable insights into the use of AI for military purposes and provide opportunities to test a wide range of combat systems supplied by Kyiv's allies and partners.[98] This is also true for Polish defense companies and the Armed Forces, whose systems have been transferred to Ukraine. The Russo-Ukrainian conflict has seen the deployment of unmanned systems supported by AI on an unprecedented scale, particularly in geospatial and imagery intelligence as well as the use of AI in BMS to support planning, in fire control systems, to monitor and detect disinformation, and data analysis[99] AI-enhanced assets help Ukraine withstand relentless Russian attacks while inflicting significant damage on Moscow's military capabilities and critical infrastructure.[100]

These developments have been closely monitored and analyzed by Polish decision-makers. The Ministry of National Defense places strong emphasis on expanding the Armed Forces' capabilities in unmanned aerial systems (UAS), acquiring significant numbers of tactical and operational drones across all branches of

94 "IDA – Inwestycje i adaptacja technologii podwójnego zastosowania" [IDA – Investments and adaptation of dual-use technologies].
95 "IDA Bootcamp 2025 – PFR szuka innowacyjnych rozwiązań dla obronności" [IDA Bootcamp 2025 – PFR seeks innovative solutions for defense].
96 "The Launch of the First Polish AI Factory – PIAST-AI."
97 "Polska ma zamiar dołączyć do programu Gigafabryk AI" [Poland intends to join the AI Gigafactory Program].
98 Bondar, Understanding the Military AI Ecosystem of Ukraine.
99 Goncharuk, Survival of the Smartest? Defense AI in Ukraine; Bendett, "Roles and Implications of AI in the Russian-Ukrainian Conflict."
100 Pardo De Santayana, "Artificial intelligence and the war in Ukraine."

the military. In late July 2025, to address the demand for relatively inexpensive, mass-produced mini-UAS and loitering munitions, the Ministry announced several key initiatives.

First, the government unveiled plans to establish a dedicated Drone Center based on the infrastructure of the Air Force Institute of Technology (Instytut Techniczny Wojsk Lotniczych, ITWL) and supported by joint scientific efforts of all military research institutes operating under the direct supervision of the Innovations Department of the Ministry of National Defense. These include Military Institute of Armored and Automotive Technology (Wojskowy Instytut Techniki Pancernej i Samochodowej, WITPiS), Military Institute of Armament Technology (Wojskowy Instytut Techniczny Uzbrojenia, WITU), Military Institute of Engineer Technology (Wojskowy Instytut Techniki Inżynieryjnej, WITI), and the Military Institute of Chemistry and Radiometry (Wojskowy Instytut Chemii i Radiometrii, WIChiR). This hub will serve as a testing ground for unmanned land, sea, and aerial systems, while also supporting their development and integration into operational service.[101] The Drone Center is expected to foster collaboration across all branches of the Armed Forces, military R&D institutes, and industry, as well as with specialized units such as the Cyberspace Defense Forces and the AI Implementation Center, given that these two entities, in particular, will play a central role in integrating AI technologies into combat platforms operated by the Armed Forces, including unmanned systems.

Second, to accelerate innovation and facilitate the deployment of unmanned systems, the Ministry has also begun establishing specialized drone laboratories within selected military units, including Poland's Special Forces and Territorial Defense Forces. These laboratories will enable the testing of technologies, tactics, and procedures related to the deployment of UAS under the program codenamed "Szerszeń" (Hornet).

Finally, to enable the mass deployment of drones, the Ministry is reportedly preparing special procurement regulations that will allow commanders to acquire certified unmanned systems or components from a ministerial list of secure and trusted suppliers under simplified procedures.[102]

101 "Rewolucja dronowa w Siłach Zbrojnych RP" [Drone Revolution in the Polish Armed Forces].
102 "Szerszeń i zielona linia, czyli zaczątek dronizacji armii" [Hornet and Green Line, the beginning of the Armed Forces' Dronization].

3.3 AI in Transformation of the Defense Industry

Considering the scale of Poland's military procurement, the need to establish sustainable mass production of combat equipment, and the government's ambition to allocate a substantial share of orders to the domestic industry, AI is emerging both as a challenge and an opportunity for Polish defense contractors. Although this issue extends beyond the direct application of AI in the military, the importance of such technologies in production automation and robotics is undeniable. According to a survey conducted by the Pulaski Foundation and KPBN, nearly 60 percent of Polish defense companies face a shortage of skilled workers—such as technicians and engineers—highlighting the growing gap between the available workforce and labor demand, driven by the country's increasingly unfavorable demographic situation.[103] At the same time, the adoption of AI in Poland's broader economy remains very low compared to other industrialized nations, with fewer than 6 percent of Polish companies using such technologies in 2024.[104] Furthermore, research by the Pulaski Foundation indicates that only 40 percent of surveyed defense companies positively assess their readiness to integrate AI into their operations.

These findings are alarming given that leading European defense companies, such as BAE Systems, are already implementing AI to boost competitiveness and enhance operational efficiency.[105] Since Poland's defense industry is dominated by large state-owned enterprises that are highly dependent on public procurement, transformation of the sector will require not only substantial funding but also profound changes in organizational culture and corporate governance. Such changes are crucial in light of the country's pressing needs in areas such as ammunition production and the manufacturing of drones and heavy equipment for the Land Forces.

The strategic documents discussed in this report omit the issue of industry transformation, as a dedicated Defense Industry Strategy is anticipated in late 2025. Nonetheless, it is clear that incorporating AI into this process will be essential. Without the ability to meet the government's requirements for both quality and delivery timelines, the Polish defense industry will be unable to fully capitalize on growing defense spending. Automation of production and other business processes will still require significant investment and knowledge transfer, which could potentially be facilitated through ongoing procurement programs, with foreign partners serving as providers of both technology and expertise.

103 "Defense and Resilience: Prospects of Companies Development."
104 Kolasa, "Innowacyjność Polski. Zestawienie – grudzień 2024" [Poland's Innovation. Report – December 2024].
105 "International Defense Insight Report, Securing strategic advantage for defense companies."

4 Organizing Defense AI

Prior to the publication of Poland's Military AI Strategy, the government's efforts aimed primarily to enhance the effectiveness of public institutions and their services, as well as to drive the digital transformation of companies across multiple industries—excluding the defense sector and the Armed Forces. However, in recent years, a more centralized and comprehensive approach has begun to take shape, involving the direct collaboration among numerous ministries. This approach has created a complex organizational framework, in which the Ministry of National Defense serves as a key actor responsible for coordinating AI-related initiatives within the defense sector.

The establishment of the Artificial Intelligence Implementation Center at the Cyberspace Defense Forces Component Command reflects this evolving dynamic and a growing recognition of the need to centralize the state's efforts in integrating AI and other advanced technologies into the Armed Forces. The decision to leverage the organizational and technological capabilities of the Cyberspace Defense Forces was deliberate. Modeled on structures used by the U.S. Armed Forces, this unit has cooperated with its American counterparts on numerous occasions, including joint exercises.[106] The Cyber Command's ability to conduct a full spectrum of defensive and offensive operations, develop innovative solutions to strengthen national cybersecurity, and maintain a proven record of collaboration with other NATO members provides a solid foundation for expanding capabilities through the integration of AI.[107] Both Polish officials[108] and foreign experts[109] emphasize that Poland's cybersecurity capabilities are among the most robust in NATO, which directly influences the nation's position in international rankings.[110]

As of mid-2025, it remains unclear when the AI Implementation Center will become fully operational. Nonetheless, the establishment of a single institution responsible for coordinating R&D projects, fostering collaboration with external partners, and overseeing implementation efforts within the Polish Armed Forces represents a significant step in the right direction. This approach is expected to enhance efficiency within the Ministry and improve interoperability among the various AI solutions introduced into the Armed Forces.

106 McGovern, "16th Air Force continues Poland Cyber Command partnership with first-ever malware attack exercise."
107 Kozłowski, "Polish "cyberclaws".
108 "AI w polskim wojsku. Szef MON zabiera głos" [AI in the Polish military: Minister of National Defence speaks out].
109 Sylvia, "European Digital Defence Priorities in an Uncertain World," pp. 18-22.
110 For instance, the National Cyber Security Index (NCSI) ranks Poland fifth globally, despite its relatively low overall level of digital development, while MIT's Cyber Defense Index 2022/23 placed Poland sixth among the top 20 countries with the most advanced cybersecurity assets. It is also worth emphasizing that the team from the Cyberspace Defense Forces has ranked among the leading participants in *Locked Shields 2025,* the annual exercise organized since 2010 by the NATO's Cooperative Cyber Defense Centre of Excellence (CCDCOE). For more, see: https://ncsi.ega.ee/ncsi-index/?order=-ncsi (last accessed 5 September 2025), "The Cyber Defense Index 2022/23;" "Polacy po raz kolejny na podium zawodów Locked Shields" [Poles once again on the podium at the Locked Shields competition].

Given that the Cyberspace Defense Forces Component Command has already developed its own battle management system and reportedly employs a range of AI-supported tools in cybersecurity, among others, the AI Implementation Center—backed by its experienced staff—should be well positioned to manage complex projects and identify commercial solutions adaptable to military requirements. Above all, the presence of such an entity within the military could ensure that AI integration is both well-coordinated and more consistent across all branches of the Armed Forces.

This centralized approach should help prevent the inefficiencies and lack of coordination seen in past procurement programs, where the Ministry of National Defense acquired multiple combat platforms with similar specifications simultaneously. While some of these acquisitions were fully justified—driven by growing tensions on NATO's eastern flank and Poland's substantial donations of military equipment to Ukraine, particularly heavy systems such as main battle tanks and artillery—in other cases, the decisions appeared questionable. They inflated logistics costs, complicated maintenance, and significantly increased life-cycle expenses. With the AI Implementation Center in place, such costly redundancies are far more likely to be avoided.

Despite the Ministry of National Defense's reluctance to provide a detailed plan for the AI Implementation Center's future operations, certain aspects of its upcoming activities have been publicly disclosed. The AI Implementation Center began recruiting specialists in mid-April 2025. According to Maj. Gen. Karol Molenda, Commander of the Cyberspace Defense Forces, one of its first priorities will be enhancing the Armed Forces' operational awareness by integrating and analyzing data from various sensors operated by the Polish Armed Forces using AI-supported systems.[111] To achieve this, the military will need to develop its own secure infrastructure to process and effectively utilize large volumes of classified data. The AI Implementation Center will also need to address numerous challenges related to data processing, storage, and oversight mechanisms, as well as establish unified procedures in the areas of data management, security, and privacy.

In May 2025, the Cyberspace Defense Forces launched the program Cyber LEGION. Its official name—Program for Consolidating the Polish Cyber Community Around the Polish Armed Forces and Cyberspace Defense Forces – Cyber LEGION—reflects its underlying concept. The initiative's primary goal is to allow cybersecurity professionals to contribute to military activities in a flexible way that does not require them to abandon their civilian careers.[112]

111 "Gen. Molenda: 14 kwietnia rusza Centrum Implementacji Sztucznej Inteligencji" [Gen. Molenda: The Artificial Intelligence Implementation Center will launch on April 14].

112 "Zostań cyberlegionistą! Wojsko rusza z nowym programem" [Become a cyber legionnaire! The military launches a new program].

5 Funding Defense AI

The true scale of Poland's defense AI spending is shrouded in secrecy, making it impossible to draw definitive conclusions from ongoing and planned modernization programs. Some of these costs are likely embedded within large-scale procurement efforts—such as those involving UAS for the Land Forces and other combat systems across all branches of the Polish Armed Forces. Given that Poland's defense spending is projected to rise from 4.7 percent of GDP in 2025 to 5 percent in 2026 and that the nation continues to devote more than half of its total military expenditure to procuring advanced weapon systems, it is likely that the budget for developing and acquiring AI technologies will also grow, even if, at present, it remains relatively modest in the context of overall military spending.[113]

As of mid-2025, it remains difficult to estimate expenditures on the acquisition of AI-enabled weapon systems. However, the Ministry of National Defense has been more forthcoming about general spending related to the structural changes in the Armed Forces required to implement scalable AI technologies. In March 2025, Cezary Tomczyk, Secretary of State in the Ministry of National Defense, announced that the Ministry had allocated PLN100M (approximately €23M) for the development of AI technologies and the establishment of the Artificial Intelligence Implementation Center within the Cyberspace Defense Forces Component Command.[114] At the same time, the Polish government is developing a plan to establish a Defense and Security Fund by fall 2025, with an estimated budget of PLN25–26bn (approximately €6bn). The fund will draw on financial resources from the EU's recovery funds[115] and will be overseen by a Coordinating Committee comprising the Ministry of Development Funds and Regional Policy, the Ministry of National Defense, the Ministry of State Assets, the Ministry of Digital Affairs, the Ministry of the Interior and Administration, the Ministry of Economic Development and Technology, and the Ministry of Infrastructure.[116] The fund's beneficiaries will include both state-owned and private companies, particularly those involved in manufacturing heavy equipment for the Armed Forces.[117]

The Coordinating Committee responsible for implementing the Fund began its work in April 2025, with one of its first tasks being to define a list of priority investments requiring financial support. Although publicly available information does not explicitly identify AI as one of the investment areas, it is worth noting that current plans include funding for dual-use infrastructure—such as cybersecurity systems for critical

113 Miłosz, "Tyle Polska wyda na obronność w 2026 roku. Idziemy na rekord" [This is how much Poland will spend on defense in 2026: we are heading for a record].
114 Palczewski, "Nowa jednostka w Wojsku Polskim. Inwestycje MON w strategiczny obszar" [A new unit in the Polish Armed Forces: Ministry of National Defence invests in a strategic area].
115 "Fundusz Bezpieczeństwa i Obronności wesprze inwestycje samorządowe" [The Security and Defense Fund will support local government investments], Ministerstwo Funduszy i Polityki Regionalnej.
116 "Środki z KPO zasilą Fundusz Bezpieczeństwa i Obronności" [Funds from the National Recovery Plan (KPO) will be allocated to the Security and Defense Fund].
117 Fundusz Bezpieczeństwa i Obronności wesprze inwestycje samorządowe" [The Security and Defense Fund will support local government investments].

infrastructure and data centers—both of which will likely involve the application of AI technologies. Preliminary plans allocate approximately PLN11–12bn (€2.6–2.8bn) to enhancing the security of dual-use infrastructure, with an additional PLN4bn (€0.94bn) earmarked for modernizing industry and supporting R&D projects in the field of dual-use technologies. This funding is also expected to support startups developing innovations for defense and security purposes.[118]

Despite the classified nature of most information, it is worth noting that Poland has numerous instruments for financing the development of AI-based dual-use technologies. These special funds are included in neither the Ministry of National Defense's budget nor the Armed Forces Support Fund, both of which provide the vast majority of financial resources for military procurement and modernization. Considering that AI is a dual-use technology—originating from R&D programs in the civilian sector and now being implemented in the military—the Polish government appears well aware of the synergistic effects that arise from developing such technologies for both civilian and defense applications. It also recognizes the opportunities for the Polish economy, particularly through the growth of domestic enterprises specializing in AI. Therefore, to maximize the overall economic impact and stimulate innovation for defense and security purposes, most initiatives aimed at creating special AI funds in Poland involve a broad range of stakeholders.

One such initiative is the ongoing establishment of the Artificial Intelligence Fund, whose operations will be overseen by the Ministry of Digital Affairs, the Ministry of National Defense, the Ministry of Science and Higher Education, the National Center for Research and Development (NCBR), the Polish Development Fund (Polski Fundusz Rozwoju, PFR), Bank Gospodarstwa Krajowego (BGK), and the National Science Center.[119] The Artificial Intelligence Fund is intended to support the development of innovative AI technologies and their future commercialization by providing expertise and assistance in business development.[120] In 2025, the Ministry of Digital Affairs plans to allocate PLN4.5bn (over €1bn) to digitalization and the development of AI and cloud services. This funding will enable the National Center for Research and Development to launch several AI-related programs, such as Ścieżka SMART, dedicated to large companies seeking cooperation with SMEs and conducting R&D projects in areas identified within the National Intelligence Specializations. The SMART program also offers additional opportunities for companies, including funding for R&D infrastructure development, digitalization, and expansion into foreign markets.[121]

118 Dmitruk, "Pierwsze środki z Funduszu Bezpieczeństwa i Obronności w 2026 roku" [The first funds from the Security and Defence Fund will be available in 2026].
119 "NCBR częścią Funduszu Sztucznej Inteligencji" [The NCBR will be part of the Artificial Intelligence Fund].
120 "PFR podpisał list intencyjny w sprawie Funduszu Sztucznej Inteligencji – nowe możliwości dla rozwoju polskiej gospodarki" [PFR Signed a Letter of Intent on the Artificial Intelligence Fund – New Opportunities for the Development of the Polish Economy].
121 "Ścieżka SMART nabór FENG.01.01-IP.01-001/24."

The National Center for Research and Development (NCBR) is also preparing the 8th edition of the INFOSTRATEG program, with a budget of PLN500M (approximately €115M), aimed at advancing AI and blockchain solutions in Poland.[122] Additionally, the Center coordinates the STEP (Strategic Technologies for Europe Platform) program—an EU financial instrument designed to support R&D initiatives across the Union—with a budget of PLN300M (€70M) allocated for deep tech development.[123] Concurrently, NCBR continues implementing the previously mentioned PERUN program, with a total budget of PLN3bn (€0.7bn) dedicated to innovations in defense and security.[124]

The list of potential sources for defense AI funding is not limited to national and EU instruments, as Poland is witnessing growing interest from private financial institutions—such as venture capital (VC) funds—in investing in innovative dual-use technologies, including AI. To streamline collaboration between venture capital funds and innovators, and to increase the likelihood of commercializing new technologies, the Polish Development Fund launched the PFR Deep Tech program with a dedicated budget of PLN300M (approximately €70M).[125] The program, developed in cooperation with the Ministry of Finance, is modeled on similar initiatives in other NATO member states, including France, the United Kingdom, and the United States.[126] The active involvement of the Polish Ministry of National Defense is anticipated at a later stage. The PFR's financial resources will be used to support joint investments with selected specialized venture capital funds in dual-use technologies such as:

- Artificial Intelligence
- Quantum Computing
- Advanced Defense Systems
- Robotics
- Space Exploration and Technology
- Broad-spectrum Cybersecurity.[127]

The total amount of funds—combining PFR's resources with those provided by participating VC funds—is estimated at PLN600M (€141M), which should be sufficient to support the development of approximately 50 to 80 innovative projects.[128]

122 Lewkowicz, "MC: 4,5 mld zł na rozwój cyfryzacji w 2025 r. w ramach Funduszu AI" [Ministry of Digital Affairs: PLN 4.5 Billion for Digitalization Development in 2025 Under the AI Fund].
123 "Platforma konkursowa" [Competition Platform], Narodowe Centrum Badań i Rozwoju.
124 "Czas na realizację projektów w zakresie badań naukowych i prac rozwojowych na rzecz obronności i bezpieczeństwa państwa – ruszył Program NCBR pk. PERUN."
125 "IDA – Inwestycje i adaptacja technologii podwójnego zastosowania."
126 Ibid.
127 Ibid.
128 "PFR Deep Tech: 600 mln PLN na inwestycje w zaawansowane technologie" [PFR Deep Tech: PLN 600 million for investments in advanced technologies].

6 Fielding and Operating Defense AI

6.1 Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance (C4ISR)

The current status of defense AI system implementation in the Polish Armed Forces remains largely unknown due to the classified nature of most information related to such plans. Little is known about the systems developed to date that have entered operational service, although certain specific applications have been publicly disclosed by the Ministry of National Defense.

The most prominent example is the previously mentioned BMS Legion, developed by the Cyberspace Defense Forces Component Command. The decision to rely on in-house expertise and programming capabilities was driven by the vulnerabilities of commercial solutions, which could be compromised by the enemy. While this approach may pose challenges in ensuring compatibility with systems used by other NATO member states, its advantages—particularly regarding system security—outweigh its drawbacks.[129] The successful development of the BMS also demonstrates the intellectual and organizational capabilities of the Cyberspace Defense Forces Component, which has succeeded in attracting highly skilled IT experts and now employs approximately 6,500 military and civilian personnel.[130] This highlights the organization's strong effectiveness, a quality particularly important for establishing and ensuring the efficient operation of the AI Implementation Center.

The target configuration of the BMS Legion is expected to significantly enhance battlefield awareness at the tactical level through the following capabilities:

1. Monitoring the positions of friendly forces;
2. Providing information on the location and movement of enemy forces;
3. Offering a dedicated tactical chat to streamline the flow of information and warnings between units in real time;
4. Supporting the decision-making process;
5. Operating sensors on combat platforms;
6. Supporting logistics of the combat platforms;
7. Facilitating the exchange of information related to the use of weapons of mass destruction (WMD).[131]

129 Molenda, Karol, Interview by Okulska, Inez, "Piąty żywioł w starciu z AI."
130 "Centrum Implementacji Sztucznej Inteligencji ma wystartować w tym roku. Polskie wojsko coraz szerzej korzysta z tej technologii" [The Artificial Intelligence Implementation Center is set to launch this year, as the Polish military increasingly adopts this technology].
131 "SI BMS Legion."

As of mid-2025, the primary function of the BMS Legion is the automated analysis of imagery data from unmanned aerial systems. Publicly available information suggests that the system is primarily intended to extend the capabilities of various platforms operated by the Land Forces[132] and to provide tactical information to JASMINE, the network centric Headquarters Management System developed by TELDAT,[133] a Polish company specializing in defense ICT systems.[134] In early January 2025, the Polish Armed Forces confirmed integrating the BMS Legion with Poland's K2 main battle tanks (MBTs).[135] This could indicate that most modern land combat platforms—including MBTs, infantry fighting vehicles (IFVs), and other heavy vehicles—will be equipped with the system in the future.

According to Maj. Gen. Karol Molenda, commander of the Polish Cyber Command, the AI Implementation Center plans to extend the application of the BMS Legion and integrate it with the Integrated Multi-Level Information System of the Ministry of National Defense (Zintegrowany Wieloszczeblowy System Informatyczny Resortu Obrony Narodowej, ZWSI RON). The ZWSI RON system comprises four dedicated modules focusing on financial resources, human resources, organizational structures, and logistics.[136] The latter module is expected to be integrated with the BMS Legion to streamline the flow of information on logistical processes and equipment within the Polish Armed Forces.[137] The Polish Armed Forces reportedly tested certain AI technologies within its battle management system for friend-or-foe identification during the Anakonda-23 exercises in 2023.[138]

6.2 Cybersecurity

Possibly one of the most advanced applications of AI in the Polish Armed Forces to date is in the field of cybersecurity. AI has reportedly been deployed by the Cyberspace Defense Forces to support the detection of anomalies and software vulnerabilities in defense ICT systems, which are particularly exposed to operations carried out by so-called APT (Advanced Persistent Threat) groups.

An APT can be defined as an attack conducted by well-organized groups using sophisticated methods to penetrate networks and gain access to specific information

132 "Za nami 'Warsztaty dotyczące wizji nowoczesnej łączności'" [The "Workshops on the Vision of Modern Communications" are behind us], Wojsko Polskie.
133 "HMS JASMINE - Headquarters Management System for Operational and Tactical Level," Teldat.
134 "Za nami 'Warsztaty dotyczące wizji nowoczesnej łączności'," Wojsko Polskie.
135 Ibid.
136 Stańczyk, "Integrated Multi-level Information System of the Ministry of National Defence as an IT System supporting the management of military resources," pp. 734-751.
137 Molenda, Karol, Interview by Okulska, Inez, "Piąty żywioł w starciu z AI," HAI Magazine.
138 Palczewski, "Wojsko RP rośnie. Generał Molenda szczerze o potencjale Polski" [The Polish Armed Forces are growing: General Molenda speaks candidly about Poland's potential].

or carry out acts of cyber sabotage.[139] Although the term originally did not specify the type of actors involved, in this context it refers to cybercriminal organizations linked to state actors—Russian special services in particular. Given the scope and scale of military and civilian ICT infrastructure and the large number of users, AI-based technologies are essential for instantly detecting potential threats to the security of the armed forces and the state's critical infrastructure.[140] Senior military officials in Poland maintain that the country is already engaged in an ongoing conflict in cyberspace, with personnel of the Polish Cyberspace Defense Forces intervening on average every two hours.[141] This is unsurprising, considering that Poland ranks third in Europe among the most targeted countries by state-sponsored cyberattacks.[142]

6.3 Other Applications

Little is known about other specific applications of AI technologies in the Polish military, although the Ministry of National Defense has acknowledged that large language models (LLMs) are used to varying degrees to automate processes, analyze images and voice data, perform real-time translations and transcriptions, and support decision-making.[143]

Beyond their roles in C4ISR and cybersecurity, one of the most likely areas for future AI application is personnel training—particularly in preparing personalized educational materials, as well as in developing scenarios and conducting data analysis for wargaming and simulations. Additionally, certain Polish-made combat platforms—such as WB Group's GLADIUS unmanned reconnaissance and strike system, which includes both reconnaissance UAVs and loitering munitions[144]—have reportedly been integrated with AI technologies. These capabilities enable the systems to operate autonomously even in the presence of hostile electronic warfare measures.[145]

The implementation of AI is also a key component of the ongoing East Shield program, which aims to fortify Poland's eastern borders with a complex system of military infrastructure, including AI-supported operations centers and automated

139 Ahmad, "Strategically-motivated advanced persistent threat: Definition, process, tactics and a disinformation model of counterattack," pp. 402-418.
140 Molenda, Karol, Interview by Okulska, Inez, "Piąty żywioł w starciu z AI," HAI Magazine.
141 Palczewski, "Ataki na polską infrastrukturę krytyczną. Dowódca DKWOC: zmieniamy podejście NATO" [Attacks on Polish critical infrastructure: Commander of the Cyberspace Defense Forces Command – We are changing NATO's approach].
142 Witcher, "Microsoft's Poland Investment: Your Next Big Break in Cybersecurity."
143 "AI w służbie MON-u" [AI in the Service of the Ministry of National Defense].
144 "First Gladius system delivered."
145 Maciejewski, "MSPO 2023: kielecka premiera systemu Gladius Grupy WB" [MSPO 2023: Kielce Premiere of WB Group's Gladius System].

combat systems.[146] Furthermore, under a contract signed in 2020, the Polish Air Force is currently expecting the delivery of 32 F-35A TR-3 Block 4, with the first aircraft scheduled to arrive in 2026.[147] The acquisition of these stealth fighter jets is perceived as a milestone in modernization and digitalization of the Polish Armed Forces[148] given their advanced armament, sensors, and the reportedly AI-enabled architecture of the TR-3 Block 4.[149]

146 "East Shield," General Staff of the Polish Armed Forces, 27 May 2024.
147 "17. polski F-35" [17th Polish F-35].
148 Palowski, "Polskie F-35: rewolucja, którą trzeba wykorzystać" [Polish F-35s: A Revolution to Capitalize On].
149 Suciu, "The F-35 Lightning II Newest Upgrade Is Almost Complete."

7 Training for Defense AI

Training of military personnel has been identified as one of the most promising applications of AI in the Polish Armed Forces, as outlined in Poland's Military AI Strategy 2024–2039. The authors of the Strategy perceive AI tools as *revolutionary*, particularly in e-learning given the possibility of personalizing materials and adapting them to the specific needs of individual soldiers.[150] The potential applications involve automating the training assessment process, creating tailored learning content, and supporting direct multilingual communication between trainers and trainees.

The Strategy also highlights the role of AI algorithms in designing scenarios and supporting the development of realistic simulations, such as those utilizing virtual reality (VR), to create more intelligent and challenging adversaries. These simulations could stimulate trainees to develop rapid decision-making skills in dynamic combat environments at the tactical, operational, and political-strategic levels, depending on the specific purpose of the training and the scenario. While such tools cannot fully replace live training, AI-supported advanced simulators can significantly reduce both the costs and risks associated with the training process.[151]

The growing importance of AI is directly influencing the operations of military universities. For example, the Military University of Technology in Warsaw (WAT) has introduced regulations governing the use of AI by both staff and students. Machine learning and AI have also been integrated into the university's IT studies curriculum.[152] At the same time, WAT actively seeks partnerships with business entities—such as IBM—to accelerate research into AI applications in fields like cybersecurity and logistics.[153]

The Ministry of National Defense is also seeking partnerships with multinational corporations not only to foster innovation in groundbreaking defense technologies, but also to advance education and research in AI, cloud technologies, and quantum computing. In February 2025 the Ministry and Microsoft agreed to expand the scope of their collaboration, particularly in cybersecurity.[154] At the same time, Microsoft signed a separate agreement with the Polish government to invest PLN2.8bn (around €0.66bn) in cloud and AI infrastructure in Poland, with completion expected by June 2026. This investment is anticipated to deliver numerous benefits to the Polish economy and workforce, including providing training

150 "Resortowa strategia sztucznej inteligencji do roku 2039," Ministerstwo Obrony Narodowej, 11 October 2024.
151 Krakowski, Krzysztof and Szewczyk, Kamil, "Systemy symulacji wirtualnej w szkoleniu bojowym żołnierzy wojsk lądowych – stan obecny i perspektywy rozwoju" [Virtual simulation systems in combat training of ground forces soldiers – current status and development perspectives].
152 "Załącznik nr 1 do uchwały Senatu WAT nr 103/WAT/2023 z dnia 22 czerwca 2023 r." [Annex No. 1 to Resolution No. 103/WAT/2023 of the Senate of the Military University of Technology of June 22, 2023], Wojskowa Akademia Techniczna – Wydział Cybernetyki, 22 June 2023.
153 "WAT zacieśnia współpracę z IBM-em" [WAT Strengthens Cooperation with IBM].
154 "Nowe technologie, prace badawczo – rozwojowe, cyberbezpieczeństwo – Wojsko Polskie rozwija współpracę z Microsoft" [New Technologies, Research and Development, Cybersecurity – The Polish Armed Forces Expand Cooperation with Microsoft].

opportunities, strengthening national cybersecurity, and supporting the education sector and other institutions with dedicated AI solutions.[155]

In September 2024, the Cybersecurity Training Center for Excellence (Eksperckie Centrum Szkolenia Cyberbezpieczeństwa, ECSC) at the Ministry of National Defense signed an agreement with Oracle to enhance the skills and competencies of the Ministry's military and civilian personnel in cyberspace, drawing on the expertise and knowledge transfer provided by the American company. The initiative's primary objective is to modernize training programs and education in military ICT infrastructure, cybersecurity, cloud computing, and AI.[156] In February 2025, the ECSC also partnered with Microsoft to incorporate Microsoft Academy courses into its training programs. This collaboration established a certified Microsoft training center, giving Ministry personnel access to cutting-edge AI and cloud solutions, as well as the opportunity to acquire the skills needed to operate them effectively in relevant operational contexts.[157]

It is also worth noting that AI and cybersecurity-related training modules and webinars have been integrated into the curriculum for Polish civil servants. For instance, in 2024, a program to train 400 civil servants included 32 hours of lectures covering available AI technologies and their practical applications, risks associated with their implementation, relevant national and international regulations—particularly the EU AI Act—as well as policies aimed at building Poland's AI ecosystem.[158]

155 "Microsoft announces a PLN 2.8 billion investment in cloud and AI infrastructure, skilling and cybersecurity in Poland."

156 "Eksperckie Centrum Szkolenia Cyberbezpieczeństwa i Oracle na rzecz modernizacji Wojska Polskiego" [Expert Cybersecurity Training Center and Oracle for the Modernization of the Polish Armed Forces], Eksperckie Centrum Szkolenia Cyberbezpieczeństwa im. prof. Zdzisława Krygowskiego, 17 September 2024.

157 "Eksperckie Centrum Szkolenia Cyberbezpieczeństwa w partnerstwie z Microsoft uruchamia akademię rozwoju kompetencji na rzecz cyberobronności kraju" [The Expert Cybersecurity Training Center, in Partnership with Microsoft, Launches a Competency Development Academy for the Country's Cyber Defense].

158 For more information on civil servants training programs, see: https://www.gov.pl/web/sluzbacywilna/plan-szkolen-centralnych-w-sluzbie-cywilnej-na-2024-r (last accessed 5 September 2025).

Table 2: Selected Cooperation Initiatives between the Polish Armed Forces and the Private Sector in Innovative Technologies

Partner	Polish Entity	Initiative/Agreement
Cisco	Cyber Defense Forces Component Command)	A partnership in the field of cybersecurity, including the exchange of knowledge on the latest cyber threats and joint efforts to strengthen the protection of critical infrastructure. Cisco's Talos unit provides information on detected threats.
Microsoft	Cyber Defense Forces Component Command and the Polish Armed Forces	DKWOC has joined the Microsoft Government Security Program (GSP), which provides controlled access to source code, threat information, and technical content. A separate cooperation agreement was signed to deepen cooperation in new technologies, including cloud computing, AI, and quantum computing, to strengthen the army's resilience.
Oracle	Cyber Defense Forces Component Command	An agreement was signed on cooperation in the field of cybersecurity, AI, and cloud infrastructure. The goal is to build joint task forces for the exchange of information on threats and the building of knowledge, which is to increase the country's cyber resilience.
Google	Cyber Defense Forces Component Command	An agreement on cooperation in the field of cybersecurity and data storage in the cloud, also including plans for joint Polish-American training. The agreement was signed for at least one year.

8 Conclusion

Assessing Poland's efforts to develop and implement AI in the defense sector is undoubtedly a difficult task, given the numerous uncertainties surrounding government plans in this area. Since the country is still at the very early stages of deploying such technologies—and with only limited details regarding this policy publicly available—a final judgment would certainly be premature. Nonetheless, the overall picture of Poland's endeavors is relatively optimistic.

The government has published a military AI strategy and has already begun implementing it, for example, by establishing the Artificial Intelligence Implementation Center. The decision to leverage the expertise and capabilities of the Cyberspace Defense Forces in launching this initiative reflects a more centralized and sound approach that could yield greater efficiency and foster synergistic effects through cooperation among all key stakeholders, including the Ministry of National Defense, the Polish Armed Forces, industry, and the R&D sector. The competence of the AI Implementation Center's staff will be crucial in this regard. However, it is equally important to note that the development of more detailed policies and regulations, as well as the creation of adaptable AI technologies for the military, will require both substantial funding and highly skilled human resources. These factors will remain significant challenges, despite Poland's growing defense budget and increasing public interest in military service.

While Poland's efforts to implement AI within the military appear relatively centralized, the same cannot be said for the broader AI ecosystem, which remains fragmented and dominated by multinational corporations. Although Polish companies—ranging from established enterprises to startups—are pursuing innovations with both civilian and military applications, Poland's reliance on foreign technology providers is undeniable. This situation is unlikely to change in the foreseeable future unless well-designed policies—accompanied by adequate funding—are introduced to support the growth of domestic technology and defense firms.

Nonetheless, it is worth noting that Poland is making efforts to stimulate the growth of the entire AI ecosystem, with particular emphasis on indigenous solutions, by establishing a range of dedicated funds and policies aimed at fostering innovation in domestic companies. These initiatives rely primarily on national resources, supplemented to some extent by EU funding, although the latter's potential could be leveraged more effectively. This is especially evident in Poland's participation in EDF projects, where Polish entities often play only a subsidiary role. It is also worth noting that opportunities to utilize such funds are recognized mainly by private companies and universities, while state-owned enterprises make use of them only to a very limited degree.

Beyond strictly military applications of AI pursued by defense companies, greater emphasis should be placed on the development of dual-use technologies. While interest in such technologies is growing and additional funds are being established to support innovative projects, it remains crucial to create an effective system that fosters business cooperation among various entities—including startups and private enterprises—to ensure the successful commercialization of new solutions. This is particularly important for building a sustainable ecosystem of domestic firms and R&D institutions, which in the long term will reduce Poland's dependence on foreign suppliers. The rationale for such policies is both economic and security driven as adopting domestic solutions could strengthen Poland's technological autonomy while stimulating the growth of local businesses. In turn, this would help reorient the economy toward more innovative, high value-added production, generating better-paid jobs, expanding the tax base, and fostering long-term economic resilience.

Promising prospects notwithstanding, the modernization and transformation of Poland's defense industry—particularly its state-owned enterprises—also remains uncertain, especially as the country continues to acquire numerous weapon systems, mostly from foreign defense contractors. The automation of production, along with the implementation of AI (for example, through smart factories), is undeniably necessary given the needs of the military and the nation's deteriorating demographic situation. While this issue lies beyond the scope of this report, further research in this area is essential to fully grasp the importance of AI in the modern defense sector, particularly in the context of the mass production of military equipment.

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